

Data Analysis Project Report: Interactive Sales Dashboard

Project Description

This project represents a data visualization mastery pipeline built using Python. By leveraging the seaborn, matplotlib, and plotly libraries, raw e-commerce data is transformed into a highly visual, interactive dashboard. The analysis focuses on understanding price distributions, regional purchase behaviors, numerical correlations, and time-series trends to empower data-driven decision-making.

Objectives

- Utilize the seaborn library for advanced statistical visualizations (density, variance, outliers).
- Generate diverse chart types (box plots, violin plots, heatmaps, subplots) to uncover hidden patterns.
- Design cohesive, professional color schemes utilizing built-in seaborn themes.
- Build an interactive, web-ready scatter plot using Plotly Express with dynamic hover data.
- Consolidate multiple static charts into a clean 2x2 subplot dashboard architecture.

Technical Requirements Implemented

- **Statistical Distributions:** Used `sns.boxplot()` and `sns.violinplot()` to visually map data density, variance, and outliers across categorical groups, providing deeper context than simple averages.
- **Correlation Heatmaps:** Implemented `sns.heatmap()` coupled with `.corr()` to mathematically prove the relationship between quantitative variables (Price, Quantity, Total Sales).
- **Grid Layouts:** Leveraged `plt.subplots(2, 2)` to seamlessly bind four distinct Seaborn charts into a single static dashboard matrix.
- **Interactive Web Visualization:** Deployed `px.scatter()` to create a dynamic, 4-dimensional chart (X, Y, Color, and Size) that can be exported as an HTML file with interactive tooltips.

Data Manipulation & Visualization Logic

The program utilizes the following advanced visualization logic to process and present the data:

Operation Performed	Python Logic Used	Business Insight / Purpose
Box Plot Summary	<code>sns.boxplot(x='Product', y='Price')</code>	Identifies median pricing, price variance, and outlier transactions across different product categories.
Density Mapping	<code>sns.violinplot(..., inner="quartile")</code>	Combines box plot features with KDE (Kernel Density Estimation) to show the exact density of quantities ordered per region.
Mathematical Correlation	<code>sns.heatmap(df.corr(), annot=True)</code>	Displays the statistical strength between numerical variables to identify the primary drivers of Total Sales.
Dashboard Grid	<code>fig, axs = plt.subplots(2, 2)</code>	Consolidates regional revenue, time-series trends, transaction volumes, and price clustering into a unified executive view.
Plotly Interactive	<code>px.scatter(..., hover_name="Region")</code>	Allows end-users to dynamically explore precise transaction details by hovering over data points on a web browser.



Setup & Installation

1. Install Python on your system
2. Install required libraries by running: `pip install pandas matplotlib seaborn plotly jupyter`
3. Download or clone this repository
4. Ensure `sales_data.csv` is placed in the root directory
5. Run the notebook using Jupyter, or execute the script using: `python dashboard.py`



Sample Input & Output

The pipeline processes raw transactional data. Here is a sample of the data being visualized:

Sales Data:

Date: 2024-01-01 | Product: Phone | Quantity: 7 | Price: 37300 | Region: East | Total_Sales: 261100

Result for Sample Data:

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✅ Data successfully loaded and dates formatted!

🎨 Generating Seaborn Statistical Visualizations...

🌟 Generating Plotly Interactive Dashboard...

✅ Saved interactive dashboard as HTML file!

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📁 Project Files

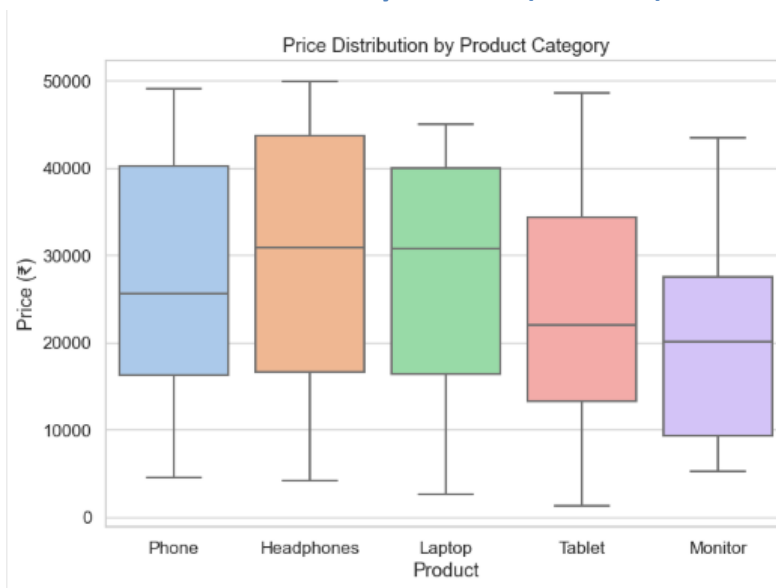
- CustomerAnalysis.ipynb – Main Python data visualization notebook
- analysis_report.docx – Final business analysis and executive summary
- sales_data.csv – Raw transactional dataset
- requirements.txt – Dependencies (pandas, matplotlib, seaborn, plotly)

📖 What I Learned

- Move beyond basic plotting to create statistical visualizations that reveal density and outliers.
- Generate mathematical correlation matrices to mathematically prove relationships between data points.
- Create complex 2x2 subplot grids to build cohesive, presentation-ready static dashboards.

📊 Business Analysis of the Graphs

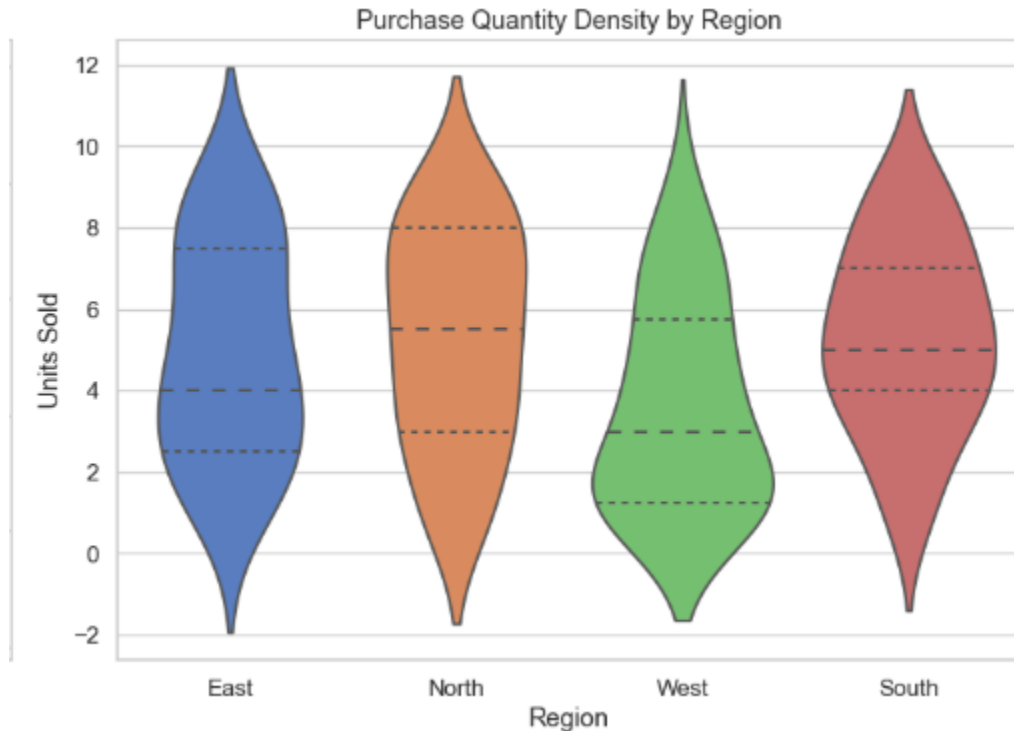
1. Price Distribution by Product (Box Plot)



Analysis: This chart reveals the pricing spread of our inventory. The "box" shows where the majority (the middle 50%) of our product prices lie. If a product category (like Laptops) has

a very "tall" box, it means we sell a vast range of them—from cheap budget models to highly expensive premium models. The dots on the outside represent "outliers," which are unusually large VIP purchases or heavily discounted clearance sales.

2. Purchase Quantity Density by Region (Violin Plot)



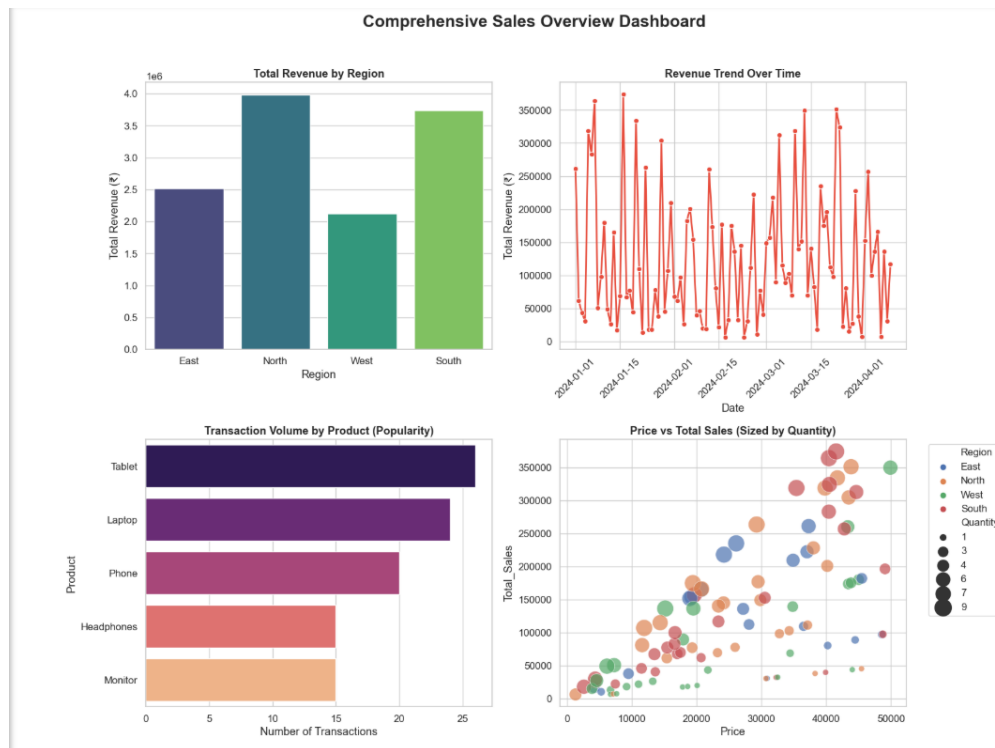
Analysis: This chart shows buying habits. If the "violin" shape is fat at the bottom (around 1-2 units), it means that region primarily consists of everyday consumers buying single items. If a region has a violin that stretches upward and gets fatter near the top (e.g., 8-10 units), it indicates that region handles our bulk/wholesale B2B buyers.

3. Correlation Matrix (Heatmap)



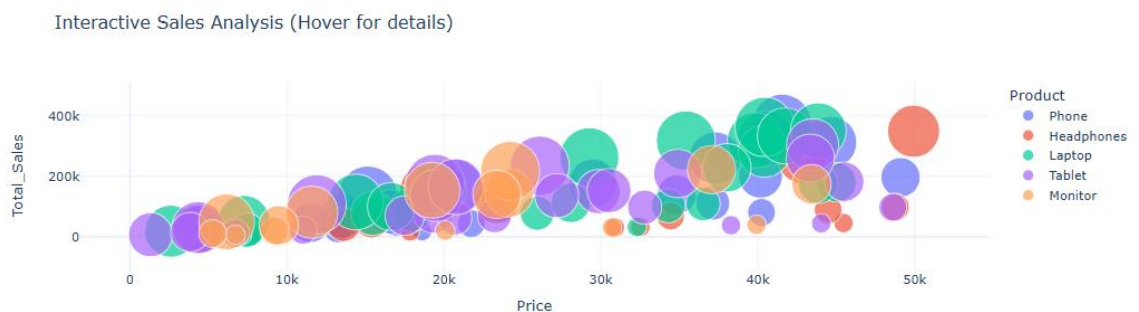
Analysis: This visually scores how closely related our numbers are from -1.0 to 1.0. Typically, you will see a strong positive correlation (closer to 1.0) between Price and Total_Sales. This is a critical business insight: it proves mathematically that pushing high-ticket items (high price) drives overall revenue significantly faster than trying to push high quantities of cheap accessories.

4. The 2x2 Static Dashboard



- **Top-Left (Bar):** Instantly highlights our most and least profitable geographical regions, showing where supply chain logistics should be focused.
- **Top-Right (Line):** Exposes seasonal revenue trends over time. Spikes and dips show us exactly when our marketing campaigns are working and when we suffer from "off-season" sales slumps.
- **Bottom-Left (Count):** Shows raw popularity. A product (like headphones) might have the highest transaction volume here, even if it doesn't generate the most revenue.
- **Bottom-Right (Scatter):** Groups our sales. We can visually identify "golden clusters"—transactions that combine both a high price tag and a large quantity.

5. Interactive Plotly Dashboard



Analysis: This is the ultimate exploration tool. By mapping Price (X), Revenue (Y), Product (Color), and Quantity (Size), we get a 4D view of our business. Management can hover over

the absolute largest bubbles in the top right quadrant to instantly see exactly which Customer ID made our most lucrative purchases, allowing for highly targeted VIP reward programs.